How much arithmetic survives a decimal expansion?

Notes accompanying the ratrecon library

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Contents

1. Setup **2**

2. The exact regime **2**
2.1 The uniqueness threshold 2
2.2 The encoding is near-optimal 3
2.3 Inversion in logarithmic time 3

3. Below the threshold: the ambiguity set **4**

4. Graph representation: the Stern–Brocot tree **5**

5. Geometric representation: the Farey tessellation **6**

6. Noise, lattices and pooling **6**

7. Constrained denominators and learned proposals **8**
7.1 Restricted denominator sets 8
7.2 Learned proposals under digit corruption 9

8. Unavoidable losses **11**

Use of AI tools **11**

References **11**

Notes on the information content of finite decimal observations of rationals. We describe what can be recovered from such observations and how efficiently, and what information is lost. The notes are expository and their theorems are classical. In particular, recovering a fraction from its leading decimal digits with the extended Euclidean algorithm is a textbook application (Shoup 2008, §4.6.1). The notes keep the question of what the digits determine separate from the question of how candidates are ranked, and they describe how the implementation in `ratrecon` *realises each bound.*

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1. Setup

Fix a height bound Q and write $\mathcal{F}_Q = \{p/q \in \mathbb{Q} : 1 \le q \le Q\}$ (numerator bounds enter trivially and are ignored here). The *observation map*

$$\rho_k : x \mapsto \text{decimal string of } x \text{ rounded to } k \text{ decimal places}$$

sends a rational to a finite string. We study it in two regimes.

1. In the *exact regime* we ask when ρ_k is injective on $\mathcal{F}_Q$, and how quickly it can be inverted.
2. In the *lossy regime* ρ_k is not injective, or the digits carry noise, and we ask what structure remains and how candidates should be ranked.

The algorithms in `ratrecon` are built on the results below, and we name the relevant routines as we go.

Experimental results quoted below are stored in `results/`, and the digit-corruption experiment of §7.2 can be reproduced by running `hybrid_gru.py`.

2. The exact regime

2.1 The uniqueness threshold

Theorem 1. Let $Q \ge 2$. If $10^k \ge Q(Q-1)$ then ρ_k is injective on $\mathcal{F}_Q$: k rounded digits determine p/q with $q \le Q$ uniquely. The same holds for truncated digits.

Proof. Let $p/q \ne p'/q'$ with $q, q' \le Q$. If $q \ne q'$ then $qq' \le Q(Q-1)$, and

$$\left|\frac{p}{q} - \frac{p'}{q'}\right| = \frac{|pq' - p'q|}{qq'} \ge \frac{1}{qq'} \ge \frac{1}{Q(Q-1)},$$

since the integer $|pq' - p'q|$ is nonzero. If $q = q'$ the two fractions differ by at least $1/q \ge 1/Q \ge 1/(Q(Q-1))$. Two reals producing the same k-digit string lie in one half-open cell of length 10^{-k}, so they are less than $10^{-k} \le 1/(Q(Q-1))$ apart. ■

The minimum gap $1/(Q(Q-1))$ is attained by $1/Q$ and $1/(Q-1)$. Zhang and Feng (2007) use the same separation to recover rationals from numerical approximations. The cruder condition $10^k \ge Q^2$, from $qq' \le Q^2$, asks for one more digit only at rare Q (below 10^6, only at $Q = 32$, 31623 and 316228). The threshold is sharp to within about half a digit. The $\frac{3}{\pi^2}Q^2 + O(Q \log Q)$ members of $\mathcal{F}_Q \cap (0, 1]$ need distinct strings, and only $10^k + 1$ strings describe values in $(0, 1]$, so injectivity requires $10^k \gtrsim \frac{3}{\pi^2}Q^2$, a factor $\pi^2/3 \approx 3.3$ below $Q(Q-1)$. `ratrecon.sufficient_digits` returns the smallest k with $10^k \ge Q(Q-1)$, and §7.1 recovers the same bound as a special case of a result on restricted denominator sets.

Relation to modular rational reconstruction. For rounded digits the problem implies a congruence. Let $p/q > 0$ and let D be the integer formed by its k rounded digits, so that the observed value is $D/10^k$. Rounding gives

$$qD - 10^k p = r, \qquad -q/2 < r \le q/2,$$

so $qD \equiv r \pmod{10^k}$ with $|r| \le Q/2$ and $0 < q \le Q$. This has the form of the congruence solved by modular rational reconstruction (Wang 1981, and Wang, Guy and Davenport 1982), with modulus 10^k, numerator bound $Q/2$ and denominator bound Q. Two fractions with the same rounded digits give pairs (r, q) and (r', q'), with r and r' in the corresponding ranges, so $q'r \equiv qr' \pmod{10^k}$ and $|q'r - qr'| < qq' \le Q^2$. If $10^k \ge Q^2$ then $r/q = r'/q'$, and $p/q = (D - r/q)/10^k$ is determined. Up to the boundary case this is the standard uniqueness condition of modular reconstruction, that twice the product of the numerator and denominator bounds be less than the modulus (Shoup 2008, Theorem 4.8). This argument gives a slightly weaker result than Theorem 1, which needs only $10^k \ge Q(Q-1)$.

The extended Euclidean algorithm that solves the modular problem, applied to 10^k and D, produces the continued-fraction expansion of $D/10^k$, which is the computation behind `cf_terms` and `convergents` (§2.3). Shoup (2008, §4.6.1) gives exactly this application for truncated digits. He recovers s/t with $0 \le s < t \le M$ from its first k digits whenever $10^k > 2M^2$, and notes that when 10^k is much smaller than M^2 the digits do not in general determine s/t, since there are $\Omega(M^2)$ such fractions.

The correspondence has limits. Wang's algorithm and the 1982 note look for a fraction u/v with $uv^{-1} \equiv c \pmod{m}$, which needs v invertible modulo m, whereas decimal denominators are often divisible by 2 or 5. For example $1/2$ gives $D = 5000$ at $k = 4$, with $(r, q) = (0, 2)$, and 2 is not invertible modulo 10^4. The congruence form used above, as in Shoup (2008), needs no such condition. The implication also runs one way only. The modular bound $|r| \le Q/2$ is looser than $-q/2 < r \le q/2$, so a modular solution need not fit the digits. For $D = 3$, $k = 2$ and $Q = 10$ the only modular solution with $|r| \le 5$ and $0 < q \le 10$ is $(r, q) = (3, 1)$, which gives $p = 0$, and no fraction with $q \le 10$ rounds to 0.03. A modular reconstruction must therefore still be checked against the observation cell, which the congruence does not record. Its output is the rounding error r/q, measured in units of the last digit, from which p/q follows as above. Truncated digits give $qD - 10^k p = -r$ with $0 \le r < q$ instead.

2.2 The encoding is near-optimal

The set $\mathcal{F}_Q \cap (0, 1]$ has $\sum_{q \le Q} \varphi(q) = \frac{3}{\pi^2} Q^2 + O(Q \log Q)$ members, so identifying one requires $\log_2(\frac{3}{\pi^2} Q^2) = 2 \log_2 Q - 1.72 + o(1)$ bits, while k digits carry $k \log_2 10$ bits. Theorem 1 needs $k \log_2 10 \ge \log_2 (Q(Q-1)) = 2 \log_2 Q - o(1)$, so the continuous separation requirement sits $\log_2(\pi^2/3) \approx 1.72$ bits above the set entropy (the half-digit gap noted in §2.1).

There is a second overhead on top of those 1.72 bits. Since k must be a whole number of digits, rounding it up adds less than a further $\log_2 10 \approx 3.32$ bits, for a worst-case total overhead of about five bits. At $Q = 100$, for example, the entropy is ≈ 11.6 bits, the continuous requirement is $\log_2 9900 \approx 13.27$ and $k = 4$ digits carry 13.29, so rounding k up costs almost nothing here. Up to these overheads, decimal digits are a near-optimal encoding of (p, q). They hide its arithmetic structure, however, and the remaining sections study the cost of recovering it.

2.3 Inversion in logarithmic time

A scan over the denominators $q \le Q$ costs $\Theta(Q)$ arithmetic steps, whereas inversion by continued fractions takes $O(k)$ arithmetic steps (on integers of $O(k)$ digits, so $O(k^2)$ bit operations with schoolbook arithmetic).

Theorem 2 (Legendre). If $|x - p/q| < 1/(2q^2)$ then p/q is a convergent of the continued fraction of x.

The true p/q lies within $10^{-k}/2$ of the centre of the observation cell, so when $q^2 < 10^k$ it satisfies Legendre's inequality with x equal to that centre. By Theorem 2, p/q is then one of the $O(\log Q)$ convergents of the centre with denominator at most Q. For rounded digits the centre is the value of the digit string itself. It is therefore enough to compute `cf_terms` followed by `convergents`, and `ratrecon` marks each candidate that passes the Legendre test (`Candidate.legendre`). The criterion certifies only membership in the convergent sequence. Several convergents can pass at once (both 22/7 and 355/113 do against π), so uniqueness of a reconstruction comes from Theorem 1 above the threshold and otherwise from exhaustive bounded enumeration (`identify`).

The interval refinement of this idea is the *Stern–Brocot descent* (`simplest_in_interval`). The observed digits pin x into an interval I, and the smallest-denominator fraction in I is found by descending the Stern–Brocot tree, one continued-fraction term per step (Richards 1981). We may speak of "the" simplest fraction because two fractions $p/q < p'/q$ with the same denominator $q \geq 2$ are never adjacent in a Farey sequence (adjacency would force $q(p' - p) = 1$), so a fraction of smaller denominator lies between them. In an interval containing at most one integer, such as a noise-free observation cell, the minimal denominator is therefore attained exactly once.

3. Below the threshold: the ambiguity set

Below the threshold the fibre $\rho_k^{-1}(\text{digits}) \cap \mathcal{F}_Q$ is the set of Farey fractions in an interval I of length 10^{-k}, and its expected size is

$$\mathbb{E}|\mathcal{F}_Q \cap I| \approx \frac{3}{\pi^2} Q^2 \cdot 10^{-k}.$$

The neighbour walk in `farey_in_interval` starts from the simplest fraction in I and enumerates this set at $O(1)$ cost per element. We treat consistency with the interval as a hard constraint, with zero likelihood outside it, and rank the surviving candidates by the score

$$\ln \text{score}(p/q) = -s \ln q - \frac{1}{2} \left(\frac{|x_{\text{obs}} - p/q|}{\sigma \cdot 10^{-k}} \right)^2,$$

with a prior $P(p/q) \propto q^{-s}$, x_{obs} the centre of the cell, and a Gaussian term of width $\sigma = 1/2 + n$ in the residual, where n is `noise_ulp`, both measured in units in the last place (ulps).

Under pure deterministic quantisation the cell already determines the likelihood completely. It is 1 for every fraction inside and 0 outside, so no residual term follows from the rounding model. The Gaussian is therefore a *centre-preference surrogate*, motivated by an analogue-noise model of the kind `noise_ulp` is designed for, in which σ grows and the cell widens. Under such a model the exact likelihood of p/q would be the noise density centred at p/q, integrated over the cell. With $s = 2$ and exact digits the residual term lies between $-1/2$ and 0, so it can reverse the order set by the prior only between denominators within a factor $e^{1/4} \approx 1.28$ of each other. With `0.856` and $q \leq 100$, for example, 83/97 (residual 0.33 ulp) outranks the simplest consistent fraction 77/90 (residual 0.44 ulp).

The default $s = 2$ is a *complexity (Occam) prior*. It penalises each candidate by its description

length $2\log q$ and so favours small denominators. It differs from the uniform distribution on the bounded Farey set. Among reduced fractions in $(0, 1]$ with $q \le Q$ there are $\varphi(q)$ with denominator exactly q, so sampling that set uniformly gives $P(q) \propto \varphi(q)$, which on average increases with q and favours large denominators. Below the threshold the prior dominates the ranking, so a benchmark there is won by whichever prior better matches the law generating the test fractions, and its top-1 accuracy does not measure the quality of the algorithm. For this reason `reconstruct` takes the exponent s as a parameter, and $s = 0$ ranks by the residual alone, closest to the observation first. For naturally occurring constants (quoted probabilities, tick prices, ratios chosen by people) small denominators are more probable, which makes $s = 2$ a sensible default.

Section 5 of `demo.py` follows $219/352$ with $Q = 400$ through $k = 3, \dots, 6$ digits. Its fibre shrinks from 49 candidates to 1 at $k = 5$, one digit before the threshold $k = 6$ of Theorem 1.

4. Graph representation: the Stern–Brocot tree

The Stern–Brocot tree gives a bijection between positive rationals and finite words in $\{L, R\}^*$, and the run-lengths of the word are the continued-fraction terms $[a_0; a_1, a_2, \dots]$ (for a rational the last run is $a_n - 1$). A rational is thus a vertex of an infinite binary tree, and every edge of the tree is an edge of the Farey graph (fractions joined when $|pq' - p'q| = 1$).

The digit interval I selects the set of tree nodes lying in I. The longest L/R prefix shared by every node in I is the part of the word that k digits determine. For any interval I of positive reals, whether closed, open or half-open, the deepest common ancestor of that set is the shallowest node f in I, which is the simplest fraction in I. The subtree below f consists of the rationals strictly between the two fractions whose mediant is f. Each of those two is a shallower node or one of the bounds $0/1$ and $1/0$, so neither lies in I, and since I is an interval containing f, every node in I descends from f. A cell that contains 0, from the digits $0.0 \dots 0$, is not an interval of positive reals, and its simplest fraction 0 is not a node. Excluded endpoints matter only for the descent itself. Run on the closure of a half-open cell, the descent can stop at the excluded endpoint (for the truncation cell $[0.4, 0.5)$ of `0.4` it would return $1/2$), so `simplest_in_interval` skips such an endpoint and returns the shallowest node inside the cell ($2/5$ in this example).

In passing from the digits to the tree we therefore keep an initial segment of the continued-fraction terms $(a_0, a_1, \dots)$ and lose the rest. The length of that segment is described by a theorem of Lochs.

Theorem 3 (Lochs, 1964). For almost every x, the number $m(k)$ of continued-fraction terms determined by the first k decimal digits satisfies

$$\frac{m(k)}{k} \longrightarrow \frac{6\ln 2 \cdot \ln 10}{\pi^2} = 0.9702\dots$$

Almost surely and asymptotically, then, one decimal digit determines just under one continued-fraction term. The two representations have nearly equal entropy rates, $\ln 10$ per digit for decimals and the entropy $\pi^2/(6\ln 2)$ of the Gauss map per term for continued fractions. The constant in Theorem 3 is their ratio, slightly below 1 because a continued-fraction term carries a little more information than a decimal digit. Rationals form a null set, so Theorem 3 says nothing about them directly. Heuristically, a rational with a large denominator behaves like a

typical real until its continued-fraction expansion terminates.

Worst cases are far from the average. Near a fraction of small height, which lies high in the tree (e.g. $x \approx 1/2$), many digits may be needed to confirm a single large continued-fraction term a_n, because the path to x is dominated by one run of length a_n. About $\log_{10} a_n$ digits already fix a depth of order a_n, while confirming a_n itself takes about $2\log_{10} a_n$, so tree depth is not a fixed multiple of the number of digits.

5. Geometric representation: the Farey tessellation

The hyperbolic plane $\mathbb{H}$ carries the Farey tessellation: ideal triangles on the vertices $\mathbb{Q} \cup \{\infty\}$, with p/q and p'/q' joined by a geodesic iff $|pq' - p'q| = 1$. This is the Farey graph of §4, drawn with geodesic edges. To each p/q we attach its *Ford circle*, the horocycle of Euclidean radius $1/(2q^2)$ tangent to $\mathbb{R}$ at p/q, so the denominator sets the geometric scale at which the rational is visible.

A real x corresponds to the vertical geodesic γ_x ending at x. The sequence of tessellation edges crossed by γ_x (its *cutting sequence*) is an L/R word whose run-lengths are again the continued-fraction terms (Series 1985). A real $x \in (0, 1)$ therefore has three representations:

- its digit string,
- its Stern–Brocot or Farey-graph path,
- the cutting sequence of a geodesic on the modular surface.

Each of them determines x. Two classical facts give a heuristic account of the rates at which information passes between them (the precise statements are Theorems 1 and 3):

- An error of ε in x moves the endpoint of γ_x by ε. The cutting sequence is unchanged until the geodesic enters Ford circles of radius $\lesssim \varepsilon$, that is, until convergent denominators reach $q_n^2 \gtrsim 1/\varepsilon$. With $\varepsilon = 10^{-k}$ this recovers the scale $q^2 \approx 10^k$ of Theorem 1, now as a statement about horoball depth.
- Denominator growth is exponential along the geodesic ($q_n^{1/n} \to e^{\pi^2/(12\ln 2)}$ a.e., Lévy 1936). The Ford circle at p_n/q_n lies at hyperbolic depth about $2\ln q_n \approx n\pi^2/(6\ln 2)$, and k digits resolve depth about $k\ln 10$. Equating the two gives the Lochs rate $n/k \to 6\ln 2 \cdot \ln 10/\pi^2$ of §4.

Passing from a finite observation to the tree or the tessellation keeps the data determined by the ε-neighbourhood of x and nothing finer, as quantified by Theorem 1 (worst case) and Theorem 3 (almost-everywhere rate). The three representations differ in which operations are efficient (comparison for digits, mediants for the tree, the $\mathrm{SL}_2(\mathbb{Z})$-action for geodesics).

6. Noise, lattices and pooling

Real observations are corrupted beyond rounding. With accuracy $|x_{\mathrm{obs}} - p/q| \le \delta$, consider the planar lattice generated by $(1, x_{\mathrm{obs}}/\delta)$ and $(0, 1/\delta)$. It contains $v = (q, (qx_{\mathrm{obs}} - p)/\delta)$ with $\|v\| \le Q\sqrt{2}$, while the Gaussian heuristic puts the shortest irrelevant vector near $\sqrt{\det}/\mathrm{const} \sim \delta^{-1/2}$. Heuristically, then, lattice reduction recovers p/q when

$$Q^2\delta \lesssim 1,$$

which at $\delta = 10^{-k}/2$ matches Theorem 1 up to a constant factor. The interval methods of §2.3 and §3 fail once the error puts the true value outside the observation cell. The lattice formulation degrades continuously in δ and generalises to several observations with a common denominator. That generalisation is simultaneous Diophantine approximation, and Lagarias (1985) analysed its complexity and its lattice-reduction algorithms.

Pooling (heuristic, with verification). For n decimals $x_1, \dots, x_n$ sharing one denominator (components of a probability vector, entries of a rational matrix row), the $(n+1)$-dimensional analogue has determinant δ^{-n} and the same Gaussian-heuristic argument predicts success when

$$Q^{n+1}\delta^n \lesssim 1.$$

Equivalently $\delta^{-1} \gtrsim Q^{1+1/n}$, against Q^2 for a single observation. The modular analogue, vector rational number reconstruction, shows a similar saving. Bright and Storjohann (2011) prove that a modulus of order $N^{1+1/c}$, for a small constant c, already confines every solution of size at most N to a lattice of dimension at most c, which they compute efficiently, whereas reconstruction one entry at a time needs a modulus of order N^2.

Ignoring constant factors and taking $\delta = 10^{-k}$, the condition reads $(n+1)\log_{10} Q \lesssim nk$, while a single observation of K digits requires $2\log_{10} Q \lesssim K$. Equating the two gives the effective precision of pooling:

$$K_{\text{eff}} = \frac{2n}{n+1}k \xrightarrow[n\to\infty]{} 2k,$$

well short of the nk digits observed in total when n is large. Pooling has diminishing returns because each new component adds one lattice dimension as well as k digits of evidence. For four 3-digit observations this gives $K_{\text{eff}} = 4.8$ digits, just more than the $2\log_{10} 200 \approx 4.6$ that a single observation needs when $Q = 200$. Section 4 of `demo.py` works through this example. With $q \le 200$ no single 3-digit decimal identifies a fraction with denominator 61, and ranked separately, each of the four decimals puts some other fraction first. Together they determine $q = 61$.

Success under these conditions is not guaranteed, since the Gaussian heuristic can fail for structured lattices. The implementation in `ratrecon.py` runs LLL (Lenstra, Lenstra and Lovász 1982) in exact arithmetic and checks every extracted candidate against the consistency bound $|qx_i - p_i| \le q\delta_i$. When no extra noise is declared, it also checks that the candidate regenerates the observed digit strings exactly. A failure of the heuristic can therefore only lower recall. The suite tests the pooled path on the examples in this section. When the lattice finds nothing and Q is at most `exhaustive_limit` (20000 by default), `reconstruct_shared` falls back to an exhaustive $O(Qn)$ scan over the observation cells, and the claim that no common denominator $q \le Q$ exists always rests on that scan. For example, with rounded digits at $Q = 100$, the LLL step misses the reconstruction $0.750, 0.293 \mapsto (3/4, 27/92)$, and the scan recovers it.

`reconstruct_shared` and `identify_shared` give three levels of guarantee, and each needs at least as much enumeration as the last.

1. On the LLL fast path the returned q is verified to be consistent, but a short lattice vector need not have the smallest first coordinate, so q may be non-minimal.

2. With `minimize=True` the returned q is the smallest feasible denominator. We scan $q = 1, 2, \ldots$ and stop at the first feasible one. Testing only the nearest numerator is enough here. Since the nearest integer minimises $|qx_i - p_i|$, if any numerator works at q then the nearest one does.
3. `identify_shared` decides whether the reconstruction is unique, and here the nearest-numerator shortcut is no longer sound. A cell of width w admits roughly $\lfloor wq \rfloor$ numerators, and as soon as $wq \geq 1$ two distinct hypotheses can share one denominator. For example, the cell $[0.95, 1.05)$ of the observation "1.0" contains both $1 = 20/20$ and $19/20$. Sampling one numerator per denominator therefore undercounts the fibre and can falsely certify uniqueness. The implementation takes the full Cartesian product of the admissible numerator ranges and counts distinct reduced tuples. A tuple represented over q and over its multiples then counts once, while two numerators at one q count twice. With a single observation `identify_shared` agrees with `identify` by construction.

7. Constrained denominators and learned proposals

7.1 Restricted denominator sets

Real sources rarely draw q uniformly. Prices move in ticks ($q \in \{2^a\}$ or $\{100, 10000\}$), probabilities are quoted in basis points, and the frequency ratios of just intonation (3/2, 5/4, 16/15) have smooth q. Restricting to an admissible set S of denominators lowers the entropy and improves the separation. Admissibility is closed under divisors: $q' \mid q \in S$ is admissible, since p'/q' is representable over q.

- Identifying an admissible fraction in a unit interval requires about $\log_2 \sum_d \varphi(d)$ bits, where d runs over the divisor closure of S (the divisors of members of S, with $d \leq Q$ when a height bound is also imposed). For $S = \{m\}$ the sum is m, the number of grid points j/m in $[0, 1)$. This compares with $2 \log_2 Q - 1.72$ bits without the restriction, so for sparse S far less information is needed.
- The separation is governed by the common grid of the denominators. Restricting to $q, q' \leq Q$ within S and reusing $1/(qq')$ would give a bound of order Q^2 again, which is no improvement. But every admissible fraction with denominator dividing some $m \in S$ lies on the grid $\frac{1}{L}\mathbb{Z}$ with $L = \operatorname{lcm}(S)$, so distinct admissible fractions are at least $1/L$ apart. For two denominators q and q' the bound sharpens to

$$\left|\frac{p}{q} - \frac{p'}{q'}\right| = \frac{|pq' - p'q|}{qq'} \geq \frac{\gcd(q, q')}{qq'} = \frac{1}{\operatorname{lcm}(q, q')},$$

with equality attained, because by Bézout $|pq' - p'q|$ ranges over the nonzero multiples of $\gcd(q, q')$. Hence the minimum spacing of the family is exactly $1/M$ with

$$M = \max_{q, q' \in S} \operatorname{lcm}(q, q'),$$

and k digits suffice as soon as $10^k \geq M$. When S is sparse, M can be far smaller than Q^2.

The maximum defining M is taken over pairs, and M can be smaller than $\operatorname{lcm}(S)$. For $S = \{6, 10, 14\}$ the total lcm is 210, but no two members reach it (the worst pair is $(10, 14)$

with lcm 70), so two digits suffice where the cruder bound would demand three. The condition $10^k \geq M$ comes from the exact spacing. It is sufficient but not always necessary, since identifiability can survive a coarser grid when the extremal pair never shares a cell. For example, $M = 110$ for $S = \{1, \dots, 11\}$, but two rounded digits already suffice. Basis points ($S = \{10^4\}$) need $k = 4$, compared with 8 for unrestricted denominators up to $Q = 10^4$, and dyadic ticks to 1/64 need $k = 2$. These bounds are computed by `minimum_spacing_denominator` and `sufficient_digits_for_denominators`.

Divisor closure also changes the enumeration. The search runs over those divisors of members of S that are at most Q, which can be found in $O\left(\sum_{m \in S} \min(\sqrt{m}, Q)\right)$ time, and then over the cell's numerators for each such denominator. This replaces the Farey walk, and for a member $m \geq Q^2$ the divisor search takes $O(Q)$ steps, independent of m. For example, $S = \{10^9\}$ with $Q = 10$ admits only $\{1, 2, 4, 5, 8, 10\}$.

Without constraints, that is for $S = \{1, \dots, Q\}$ with $Q \geq 2$, the largest pairwise lcm is $\operatorname{lcm}(Q, Q-1) = Q(Q-1)$, because $\operatorname{lcm}(q, q') \leq qq' \leq Q(Q-1)$ when $q \neq q'$ and $\operatorname{lcm}(q, q) = q$. The condition $10^k \geq M$ is then Theorem 1. `ratrecon.sufficient_digits` implements it, so `sufficient_digits(100) = 4` and `sufficient_digits(32) = 3`, where $10^k \geq Q^2$ would give 4. The test suite verifies both values exhaustively. All 3043 proper reduced fractions with $q \leq 100$ are recovered exactly at $k = 4$, and all 323 with $q \leq 32$ at $k = 3$, in both rounding and truncation modes, and one digit fewer leaves some fraction ambiguous in each case. Particular fractions can need fewer digits. At $Q = 100$, for example, 0.333 already determines 1/3.

7.2 Learned proposals under digit corruption

Above the threshold clean digits determine p/q (§2.1), but a corrupted digit usually moves the observation into a cell that holds no fraction with $q \leq Q$, and occasionally into the cell of another fraction. A sequence model (a GRU) trained on corrupted digits can propose denominators for such inputs, in a *proposer–verifier* scheme that we call the hybrid. The network proposes a shortlist of grid denominators N, and each proposal covers every reduced denominator dividing N. The fractions over those grids that are consistent with the observation are then selected in exact arithmetic and ranked by the score of §3.

The experiment in `hybrid_gru.py`, whose docstring gives the full protocol, tests this scheme. The fractions are all $p/q \in (0, 1)$ with $q \leq 100$, divided 80/10/10 into training, validation and test sets with no fraction in more than one set, and each is observed through $k = 12$ rounded digits. Theorem 1 needs only $k = 4$ here, so the experiment runs eight digits above the threshold. Clean inputs are never ambiguous, and the task is to correct digit errors rather than to rank an ambiguous fibre. Each digit is replaced with probability 0.1 by a uniformly random digit, which can equal the original, so a digit actually changes with probability 0.09.

The table gives exact-fraction top-1 accuracy as mean ± sample standard deviation over three seeds on one fixed split. The seed sets the network initialisation and the corruptions, and within a seed every method is scored on the same 1525 corrupted inputs (five corruptions of each of the 305 test fractions). The per-seed scores are stored in `results/`, together with the counts quoted below.

method	top-1
classical baseline, prior $s = 2$	0.685 ± 0.008
hybrid, $s = 2$	0.703 ± 0.007
classical baseline, $s = 0$	0.691 ± 0.004
hybrid, $s = 0$	0.709 ± 0.009
hybrid given only the true denominator, $s = 2$	0.798 ± 0.011
hybrid given only the true denominator, $s = 0$	0.817 ± 0.011
nearest codeword, ties to the smallest denominator	0.945 ± 0.004
nearest codeword, every tie counted as a miss	0.920 ± 0.001

On clean inputs the classical baseline scores 1.000 and the hybrid 0.997. At $s = 2$ the hybrid's gain over the classical baseline is 3.4, 1.3 and 0.6 percentage points in the three seeds. At the denominator level the hybrid reaches 0.725 ± 0.009 against 0.694 ± 0.006 for the classical baseline, and the GRU alone 0.490 ± 0.002.

Under this corruption the probability of the observed string, given a candidate whose 12 digits differ from it in d places, is $0.91^{12-d} \cdot 0.01^{d}$, which decreases with d. The test fractions are drawn uniformly, so the maximum a posteriori decoder returns the candidate at the smallest Hamming distance, at a cost of 3043×12 digit comparisons per input. About 5% of inputs have more than one nearest candidate, which is why the tie rule moves the score. This decoder recovers 94.5% of the fractions, against 70.3% for the hybrid.

The classical baseline falls short because its noise model is wrong for this corruption. It treats corruption as a shift in value. Its slack ladder widens the cell step by step, up to 2×10^{11} ulps (± 0.2), until candidates appear, and the baseline then ranks them by the prior and by distance to the corrupted value. A changed digit in place j moves the value by up to $9 \cdot 10^{-j}$, so the cell must widen to about that size, and the correct digits after place j then carry almost no weight. Pooled over the three seeds there are 4575 test inputs, and 1169 of them have a changed digit in the first three places. The classical baseline recovers 1 of these. The nearest-codeword decoder recovers 918, or 804 with every tie counted as a miss.

Three further points concern the hybrid.

1. With the prior that matches the uniform draw of test fractions ($s = 0$ for both methods) the classical baseline scores 0.691 and the hybrid 0.709, so the hybrid's gain of about 1.8 percentage points over the classical baseline does not come from the choice of prior.
2. The hybrid's exact check is made against the widened cell, which can extend to ± 0.2 around the observed value. It certifies only that the returned fraction lies in that cell, so under digit substitution the exact verification available for clean digits does not carry over.
3. Given only the true denominator, the hybrid recovers 79.8% at $s = 2$ and 81.7% at $s = 0$. It loses the rest because the widened cell either misses the true fraction or contains other fractions that rank above it. At $s = 2$ any fraction in the cell whose denominator is a proper divisor of q ranks above p/q. The proposal is exact here, so every miss comes from the verifier.

In this pilot the noise model matters more than the proposer. At $q \leq 100$ exhaustive nearest-codeword decoding is cheap and optimal under a uniform prior, and it beats the hybrid by about 24 percentage points, so for the learned component the experiment is a negative result.

Learned proposals might help where the noise model is unknown, or where Q is large enough that exhaustive decoding over about $\frac{3}{\pi^2}Q^2$ candidates is infeasible. In the second case the exact posterior, computed at a smaller Q where enumeration is feasible, would give a Bayes reference to measure them against. The experiment uses one architecture, one denominator bound and one noise rate. Three seeds on one data split are too few to establish the difference between the hybrid and the classical baseline statistically, but the gap to the nearest-codeword decoder is far larger than the spread across seeds.

8. Unavoidable losses

1. *Cardinality.* Below the threshold, ρ_k is on average $\sim \frac{3}{\pi^2}Q^2 10^{-k}$-to-one, and the digits alone cannot determine p/q. Only a prior (§3) or extra structure (§6, §7.1) can help.
2. *Discontinuity.* The denominator map $p/q \mapsto q$ is unbounded on every interval, so no continuous function of the observed value can compute it globally, and an exact reconstructor over unbounded height is discontinuous everywhere. Restricted to $q \le Q$ and finite digit strings, computing the map becomes a finite classification problem, and a network can in principle represent the resulting classifier. There is then no topological obstruction, and the difficulty that remains is generalisation to unseen digit sequences, which is the setting of the experiment in `hybrid_gru.py` (§7.2).
3. *Certification gaps.* Determining p/q needs $O(\log q)$ digits, but even seeing its period twice needs $s+2t \le k$, where s is the length of the preperiod and $t = \mathrm{ord}_{\tilde{q}}(10)$, with $\tilde{q}$ the part of q prime to 10. The period t can be as large as $q-1$ (full-reptend primes 7, 17, 19, 23, 29, 47, 59, 61, 97, ...), exponentially more digits than the value itself requires, a comparison Shoup (2008, §4.6.1) also makes. Before proposing a candidate, `periodic_candidates` checks that the period appears twice. No finite window can certify that the unseen continuation stays periodic, so these proposals are verified only against the window.
4. *Change of representation.* Without a height bound, the first k digits fix only an initial segment of the continued-fraction expansion (§4). For almost every real x it has about $0.97k$ terms when k is large (Theorem 3), and near a fraction of small height it can be much shorter. With the bound $q \le Q$ and k above the threshold nothing is lost, since the digits then determine p/q and with it every term.

Use of AI tools

AI tools were used for programming and for checking computations.

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